Cambridge Dynamics

Design/Strategy Review

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UR5 - Software: Task Ordering

Task Selection

- Priority Queue Data Structure
- Weigh each task by (# of points) * (stat. of dependencies)
    - Order of importance
- Operator input for task selection
- Adjust a weight to 0 to stop trying a task (for example, if points have been acquired or the logic is faulty)

Preliminary Roadmap

Heat Plate -> Place on Tray -> Pour Drink -> Place on Tray -> Heat Bowl -> Place on Tray -> Push Tray to Meal Ready Area -> Signal “Ready” to Mobile Robot

Task Breakdown

Opening Microwave:
Grasp vertical handle (1) -> provide **counterforce** next to door (2) -> **pull** along an **arc** (1)
Place Item in Microwave:
Grab and **lift** item (2) -> **place** on microwave plate (2)
Closing Microwave:
Grasp vertical handle (1) -> **push** along an **arc** (1)
Remove Item from Microwave:
Grab and **lift** item (2) -> move item to tray (2) -> **place** on tray (2)
Pouring Drink:
Grab bottle (1) -> **grab** cup (2) -> **lift** and **tilt** bottle to **pour** into cup (1) -> **move** cup to tray (2) -> **place** on tray (2)

If time permits: Stir Drink with a Spoon, Bimanual Manipulation

UR5 - Software: Robot and Operator Collaboration

Communication Options (Mobile Robot and UR5)

- Run both on the same program - not ideal, ideally should be decoupled
- HTTP server - a simple option, but less featureful
- ROS connectivity - more complex, but can tolerate more use cases

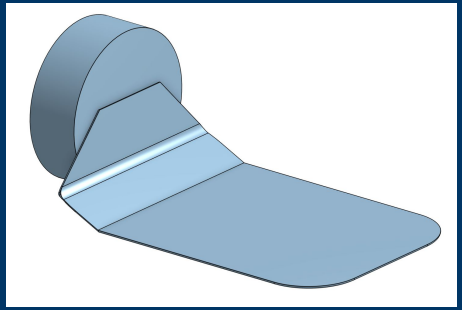
Key Strategies

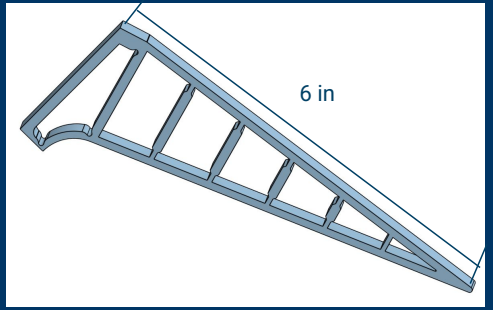
- Computer Vision for tray detection
- Operator task selection once the tray is ready

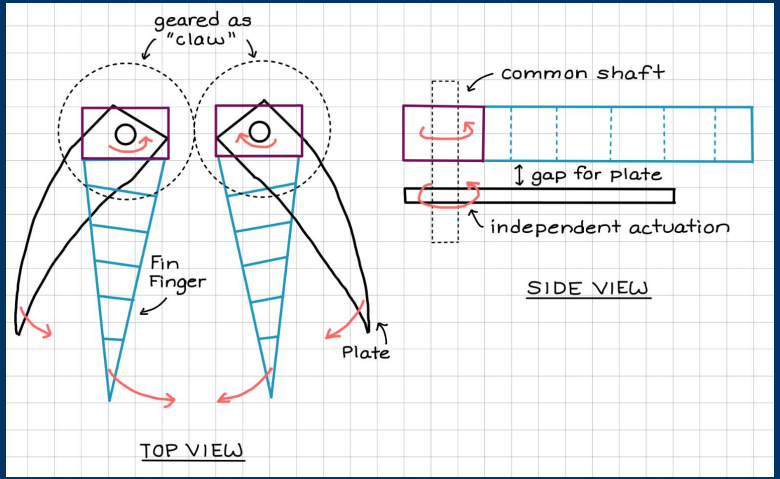
Robot Operator

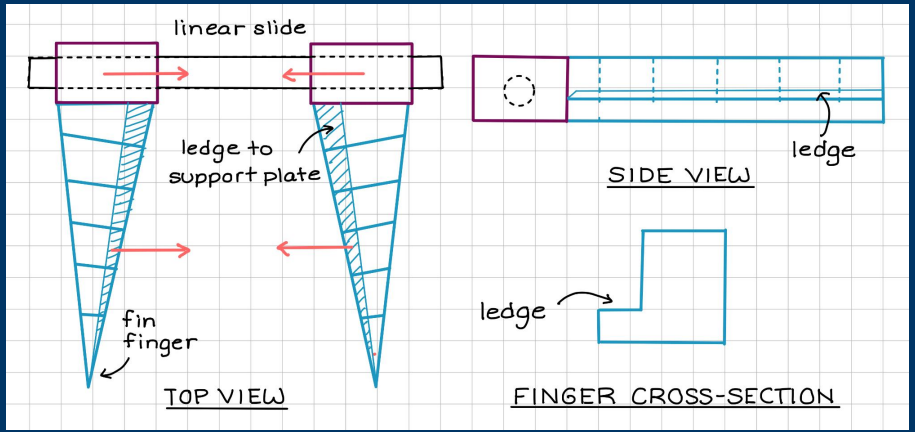
- Break down tasks into subtasks
- Operator chooses next subtask
- Emergency stop for large issues
- Mode to switch to manual control

UR5 - Hardware









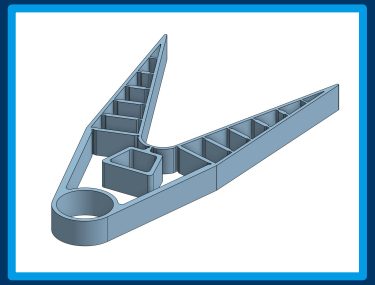
Distribution of Tasks

Arm 1: Manipulation of cup, bowl, and plate – large surface area, curvature >> **Fully custom gripper**

Arm 2: Manipulation of bottle, spoon, and microwave door handle - greater force concentrated in smaller surface area >> **Default gripper w/ compliant attachments**

UR5 - Hardware: Video Examples

UR5 - Hardware: Design Considerations



Advantages
- Compliance and dexterity provided by each fin while exerting clamping force
- Ability to grip objects with thin lips, and a variety of shapes like microwave handle

Drawbacks/predicted challenges
- Controlling actuator movement to apply the correct amount of force
- Limited normal force can be applied to hold object (weight considerations)

Fabrication and Actuation Method
- 3D Print (compliant material)
- Linear actuator to slide fins together or use motors to create “pinching” by rotating fins inward

Advantages
- Provides stability to complement the compliant fin gripper
- Simpler design for bimanual coordination
- Robust and able to bear force and weight

Drawbacks/predicted challenges
- Controlling orientation and position of spatula to ensure object is secured
- Risk of just pushing the object around

Fabrication and Actuation Method
- 3D Print (rigid material) or sheet metal
- Motor

Mobile - Software: Automation Strategy

Behavior-based autonomy

- Build a tree of modes that the vehicle could be in:
  - Go to prep area
  - Retrieve tray
  - Go to table 1
  - Deposit tray
  - Go to table 2
  - Retrieve tray
  - Go to dishwasher
- Each mode could have submodes
- Modes are determined by set of logic and state conditions
- Allows us to design behaviors for tasks independently

Mobile - Software: Considerations

State Estimation/Localization

- AprilTags for relative pose estimation to fixed known landmarks
- LIDAR/CV for object detection (dishwasher, ramp, tables, child)

Path Planning

- RRT for online planning
- Could potentially include vehicle dynamics to inform reasonable paths
- Min-jerk trajectories to reduce food movement

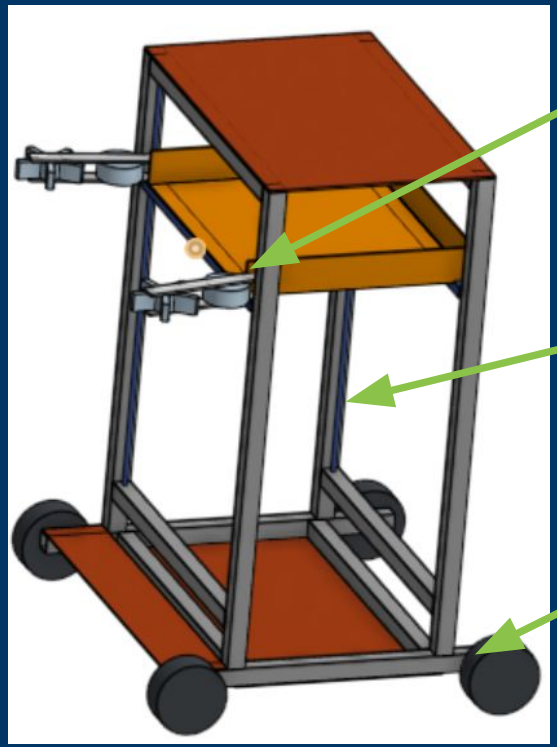
Control

- Control for tray handling: includes intake onto platform, elevator movement, and depositing to table/dishwasher
- Vehicle control: position, speed regulation

Collision Avoidance

- CV for detection of moving obstacle
- Estimate obstacle velocity by measuring distance traveled across subsequent frames

Mobile - Hardware: Designs



Task 3: Manipulating Trays

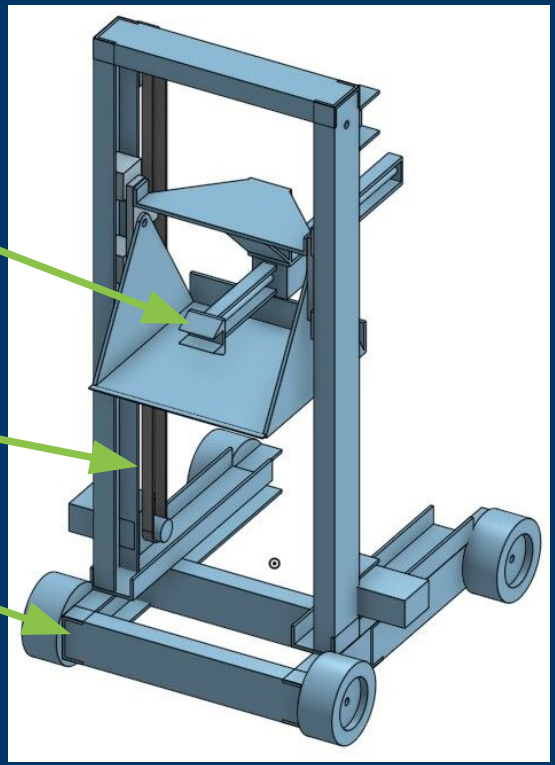
- Roller
- Prismatic arm

Task 2: Matching Height

- Lead screw + linear actuator
- Pulley elevator

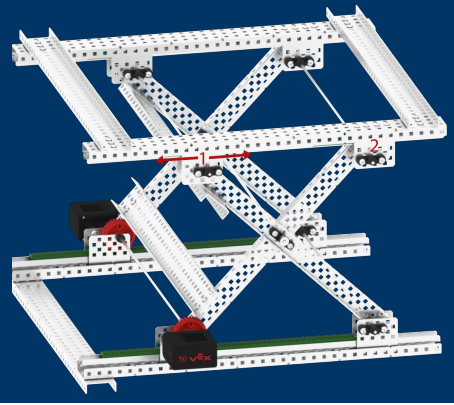
Task 1: Navigating the Field

- Base platform + camera + LIDAR



Mobile - Hardware: Design Considerations

Alternative mechanism



Outstanding questions

- Ramp traversing - would speed reduction work?
- Are tray size and location of tray on table fixed?

Implementation Detail

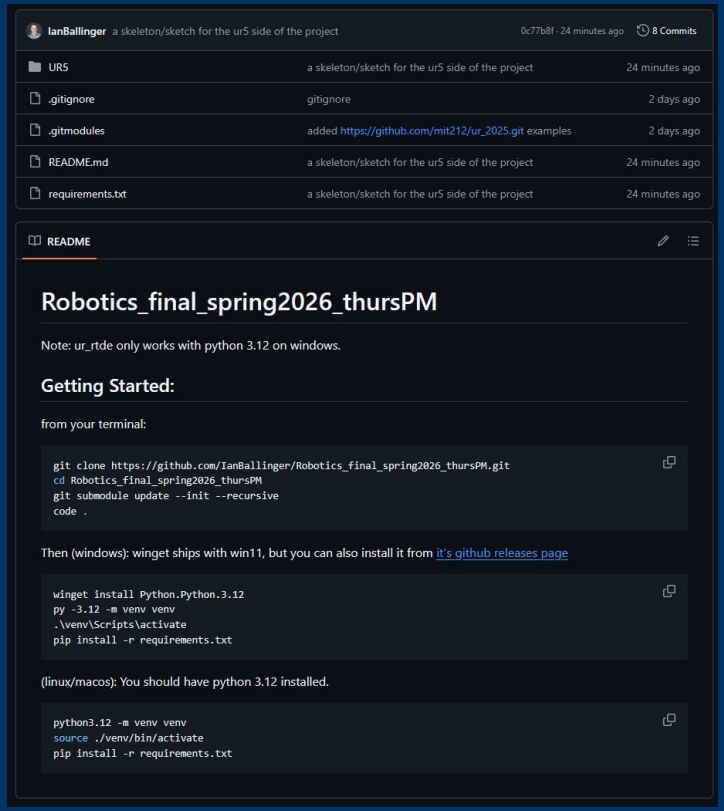
- Mobile base: default provided
- Structure/frame: aluminum
- Electronics/control: force sensor, LIDAR, camera
- Elevator:
  - If lead screw: lead screw, lead nut, stepper motor, shaft coupler, linear rails, bearing supports
  - If pulley: DC gearmotor, winch drum, cable, pulleys, linear rails, cable clamps, tensioner
- Grabbing mechanism: 3D printed parts, linear actuator

Backup Slides

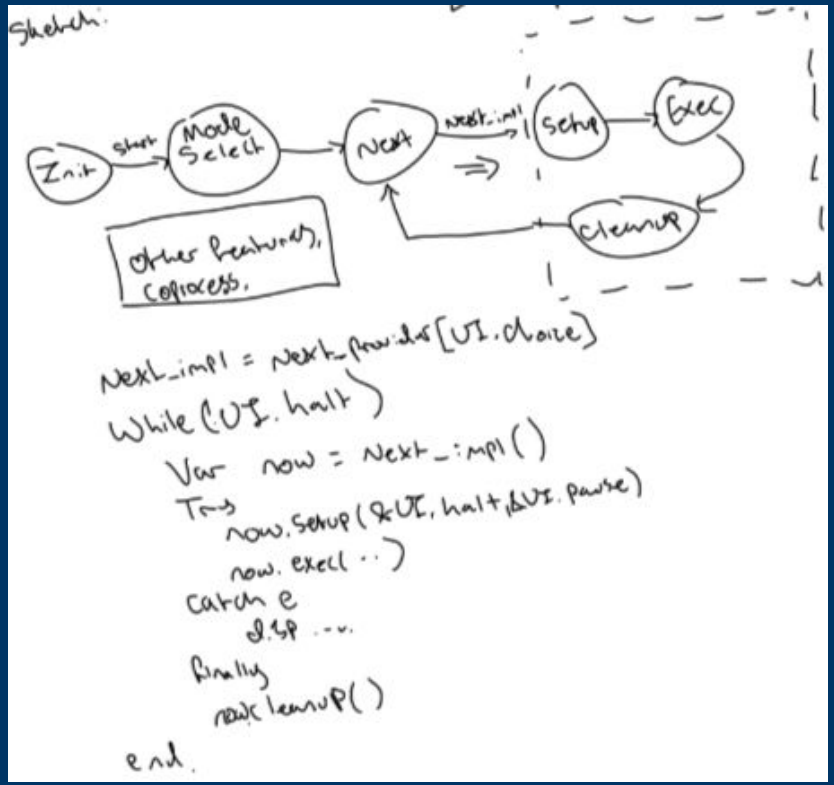
Github Repository

Follow along with our progress!

https://github.com/IanBallinger/
Robotics_final_spring2026_thursPM



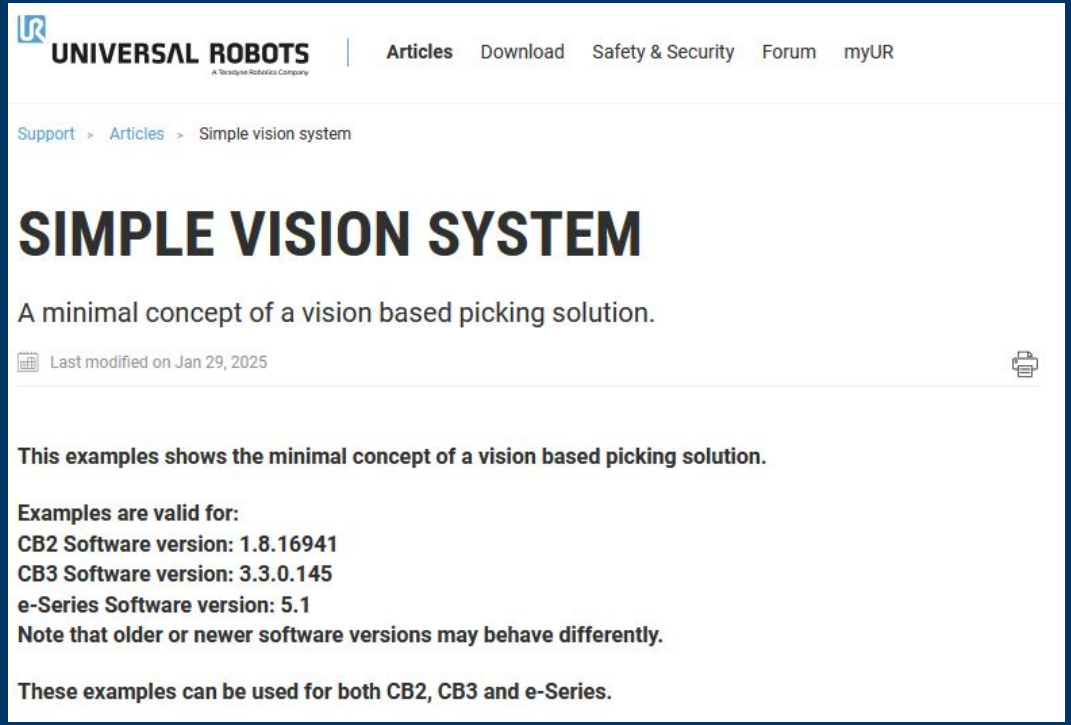
UR5 - User Input and Task Encapsulation




```python
if __name__ == "__main__":
    # Example usage
    class ExampleTask(UR5TaskInterface):
        """small example of UR5TaskInterface usage"""
        def setup(self):
            return True
        def perform_task_logic(self):
            return False
        def cleanup(self):
            pass
    task = ExampleTask(robot_ip="192.168.0.1")
    try:
        if not task.setup():
            raise RuntimeError("broken setup step")
        if not task.execute():
            raise RuntimeError("broken execute step")

    except RuntimeError as error:
        print('Caught this error: ' + repr(error))

    finally:
        task.cleanup()

    sys.exit(0)
```

UR5 - Initial Item Localization

We will use the knowledgebase articles provided by UR to bootstrap this effort. (specifically, guidance on integrating vision systems into the robot controller)



https://www.universal-robots.com/articles/ur/programming/simple-vision-system/

UR5 - Elaboration on Priority Queue Data Structure

Task selection: Priority queue data structure (the canonical solution to this problem)

Key idea: Weight each task by (the number of points it enables) * (the status of its dependencies), and then do everything in order of importance.

Initial task list:[**heat plate**, heat bowl, pour drink, signal-ready to mobile bot]

Depends on: [(**available tray**, e-stop), (available tray, e-stop), (available tray, e-stop), (drink + bowl + plate)]

Task weights: [6*1, 6*1, 6*1, 4*0]

Each arm can **pick it's own task and run independently** based on hardware ability, tasks can be **collaborative**

Can be adjusted on the fly (i.e. there is a problem, adjust a weight to 0 to stop trying a task until it's fixed then change the number back) => a natural way for operator input to select tasks

From here: subtasks (arm1 movement to open microwave, arm2+arm1 movement to grab plate, …)

Mobile - Software Requirements

- Largely similar to UR5
- OpenCV and AprilTag library for vision and pose estimation
- Existing behavior tree libraries like py_trees for the higher-level autonomy framework
- RTDE for intra-robot comms and comms to operator
- Control and planning are custom
    - Allows us to design and tune specifically for our platform specs

Mobile - Hardware: Chain lift

Mobile - Hardware: Prismatic Arm